

# Resistive Potential Divider

ENGS124 Electricity and Magnetism Lab

7<sup>th</sup> October, 2025

## Part 1

### Introduction

When several resistors are connected in series, the current flowing through each resistor is the same. Turns out the multi meter when we use to measure something in the circuit has its own resistance as it behaves as a resistor itself when connected in series. An experiment is conducted to check if it is worth considering its resistance into account when measuring something.

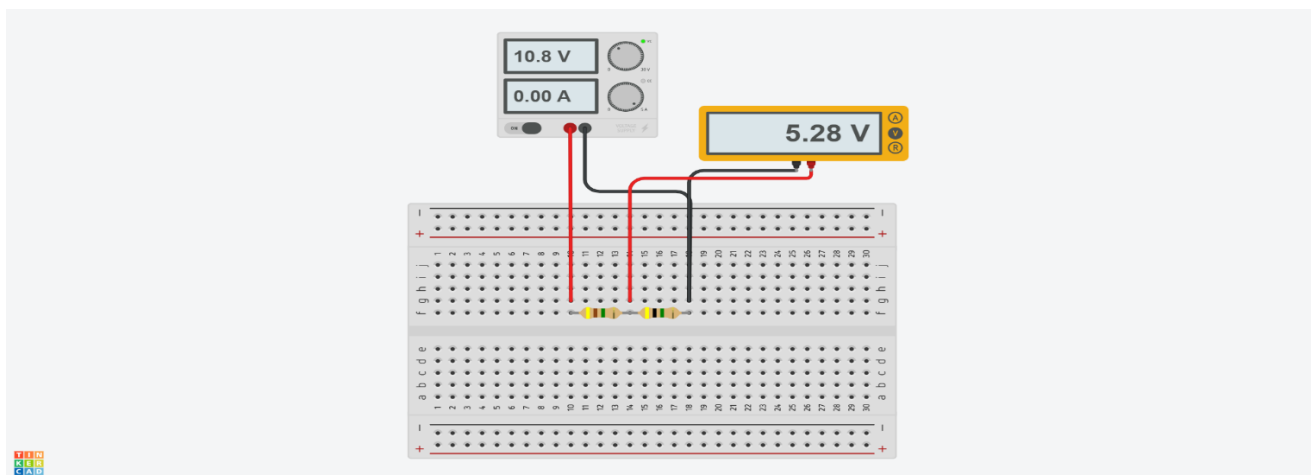
$$I = \frac{V_{\text{total}}}{R_1 + R_2}$$

$$R_2 = \frac{R_2 R_v}{R_2 + R_v}$$

$$V_2 = \frac{R_2 R_v}{R_2 + R_v} \cdot \frac{V_{\text{all}}}{R_1 + \frac{R_2 R_v}{R_2 + R_v}}$$

$$R_v = \frac{\frac{V_2 R_1}{V_{\text{all}} - V_2} \cdot R_2}{R_2 - \frac{V_2 R_1}{V_{\text{all}} - V_2}} = \frac{V_2 R_1 R_2}{(V_{\text{all}} - V_2) R_2 - V_2 R_1}$$

### Setup



## Instruments

| Name             | Quantity | Properties                                                                              |
|------------------|----------|-----------------------------------------------------------------------------------------|
| Breadboard       | 1        | Board for connecting electrical components                                              |
| Resistors        | 2        | Resistors with various resistances (Resolution 0.01 M $\Omega$ )                        |
| Multi Meter      | 1        | With two 6V and 60V volt metering modes. (Corresponding resolutions 0.001 V and 0.01 V) |
| Power Supply 15V | 1        | Generate AC/DC current through the circuit.                                             |
| Jumper Wires     | Many     | Connection Wires                                                                        |

## Variables

| Name      | Type      | Resolution |
|-----------|-----------|------------|
| $R_1$     | Parameter | 1 $\Omega$ |
| $R_2$     | Parameter | 1 $\Omega$ |
| $R_V$     | Parameter | 1 $\Omega$ |
| $v_1$     | Variable  | 0.01 V     |
| $v_2$     | Variable  | 0.01 V     |
| $v_{all}$ | Control   | 0.01 V     |
| I         | Control   | 1 A        |

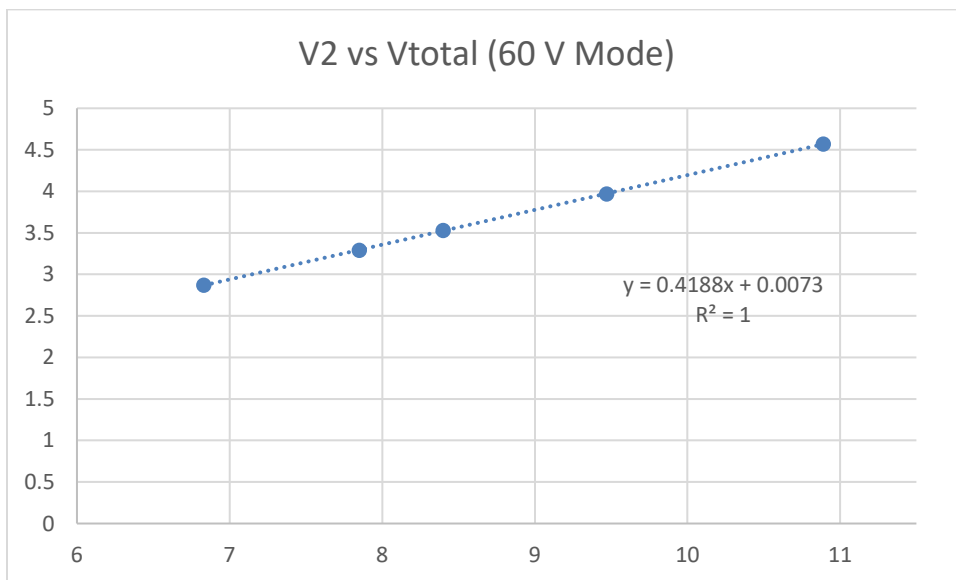
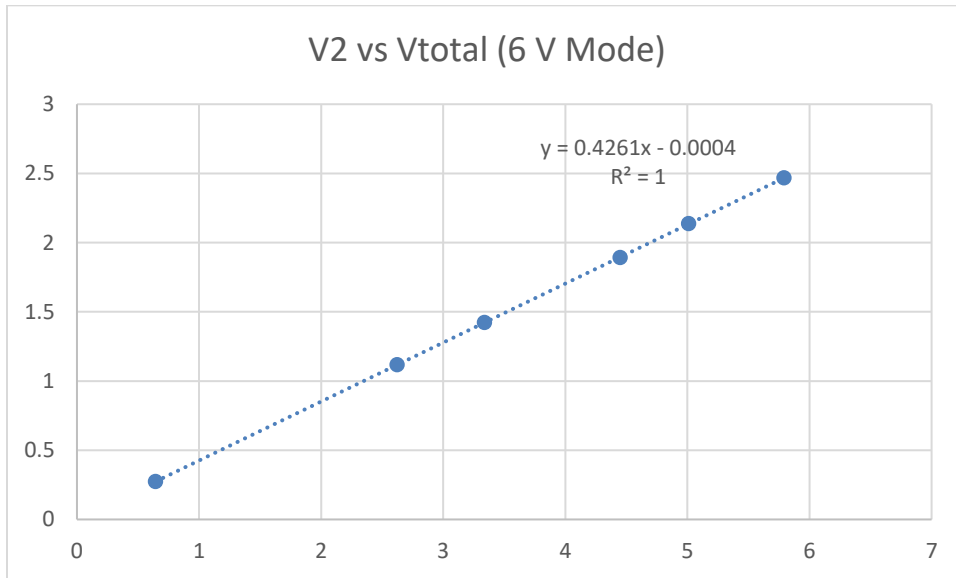
## Errors

| Name                                        | Type       | Solution                                                                       |
|---------------------------------------------|------------|--------------------------------------------------------------------------------|
| Resistance within wires and the multi meter | Systematic | Take into account                                                              |
| Voltmeter dash screen value oscillations    | Random     | Try to hold still the terminals or just change the multi meter to a better one |
| Bad contact between the components          | Random     | Try to make the connections more firm                                          |

## Data

| $V_{total}$ (V) | $V_2$ (V) | Mode | $R_1$ (M $\Omega$ ) | $R_2$ (M $\Omega$ ) | $R_V$ (M $\Omega$ ) |
|-----------------|-----------|------|---------------------|---------------------|---------------------|
| 0.644           | 0.274     | 6 V  | 4.05                | 4.001               | 11.978              |
| 2.623           | 1.118     | 6 V  | 4.05                | 4.001               | 12.129              |
| 3.338           | 1.422     | 6 V  | 4.05                | 4.001               | 12.084              |

|       |       |      |      |       |        |
|-------|-------|------|------|-------|--------|
| 4.448 | 1.892 | 6 V  | 4.05 | 4.001 | 11.957 |
| 5.011 | 2.136 | 6 V  | 4.05 | 4.001 | 12.136 |
| 5.792 | 2.468 | 6 V  | 4.05 | 4.001 | 12.104 |
| 6.83  | 2.87  | 60 V | 4.05 | 4.001 | 11.02  |
| 7.85  | 3.29  | 60 v | 4.05 | 4.001 | 10.84  |
| 8.4   | 3.53  | 60 V | 4.05 | 4.001 | 11.02  |
| 9.47  | 3.97  | 60 V | 4.05 | 4.001 | 10.85  |
| 10.89 | 4.57  | 60 V | 4.05 | 4.001 | 10.93  |



Average for 6 V Mode  $R_V = 12.065 \text{ M}\Omega$ , Standard Deviation of  $0.078 \text{ M}\Omega$

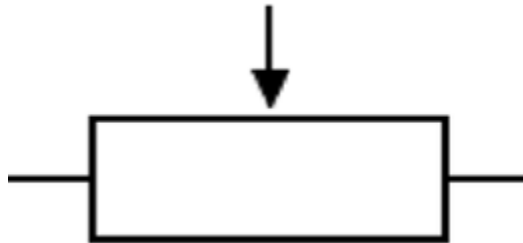
Average for 60 V Mode  $R_V = 10.93 \text{ M}\Omega$ , Standard Deviation of  $0.09 \text{ M}\Omega$

## Part 2

### Introduction

The goal of this experiment is to show how a rheostat works. The expression for the voltage drop across that resistor is.

$$V_{all} = \frac{R_2}{R_1 + R_2}$$



### Errors

| Name                                     | Type       | Solution                                                                       |
|------------------------------------------|------------|--------------------------------------------------------------------------------|
| Resistance within wires                  | Systematic | Use wires with material that has low resistance                                |
| Voltmeter dash screen value oscillations | Random     | Try to hold still the terminals or just change the multi meter to a better one |
| Poor connections                         | Random     | Make connections more stable                                                   |

## Variables

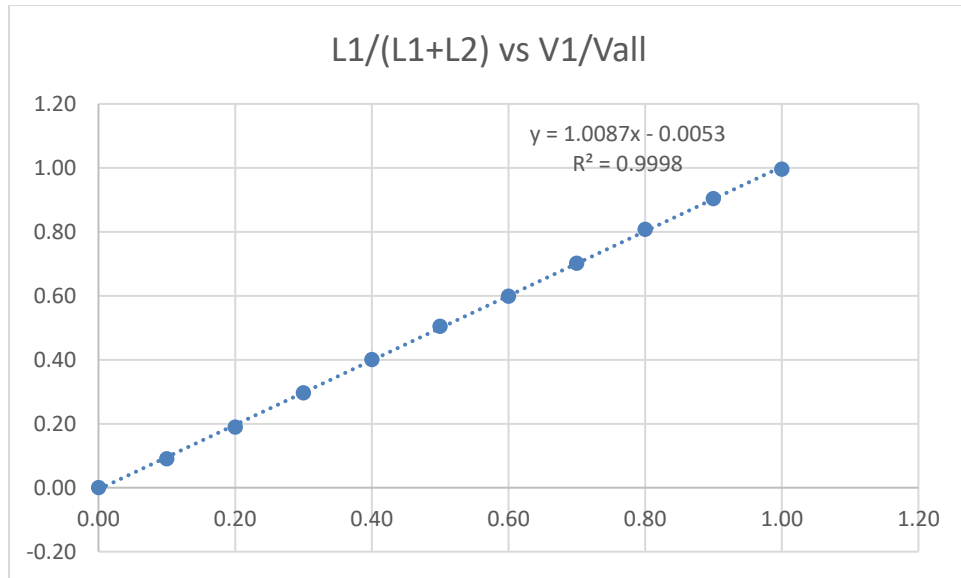
| Name      | Type      | Resolution |
|-----------|-----------|------------|
| $L_1$     | Control   | 0.001 m    |
| $L_2$     | Variable  | 0.001 m    |
| $V_1$     | Measure   | 0.001 V    |
| $V_{all}$ | Parameter | 0.01 V     |

## Instruments

| Name             | Quantity | Properties                                                     |
|------------------|----------|----------------------------------------------------------------|
| Rheostat         | 1        | Coil that lets you adjust the length and change the resistance |
| Alligator Clips  | ~        | Type of wire with alligator interface                          |
| Multi Meter      | 1        | Resolution of 0.001 V                                          |
| Power Supply 15V | 1        | Generate AC/DC current through the circuit.                    |

## Data

| L1   | L2   | V1    | Vall  | $L1/(L1+L2)$ | $V1/Vall$ |
|------|------|-------|-------|--------------|-----------|
| 0    | 0.3  | 0     | 4.766 | 0.00         | 0.00      |
| 0.03 | 0.27 | 0.43  | 4.766 | 0.10         | 0.09      |
| 0.06 | 0.24 | 0.905 | 4.766 | 0.20         | 0.19      |
| 0.09 | 0.21 | 1.412 | 4.766 | 0.30         | 0.30      |
| 0.12 | 0.18 | 1.911 | 4.766 | 0.40         | 0.40      |
| 0.15 | 0.15 | 2.404 | 4.766 | 0.50         | 0.50      |
| 0.18 | 0.12 | 2.855 | 4.766 | 0.60         | 0.60      |
| 0.21 | 0.09 | 3.344 | 4.766 | 0.70         | 0.70      |
| 0.24 | 0.06 | 3.848 | 4.766 | 0.80         | 0.81      |
| 0.27 | 0.03 | 4.309 | 4.766 | 0.90         | 0.90      |
| 0.3  | 0    | 4.746 | 4.766 | 1.00         | 1.00      |



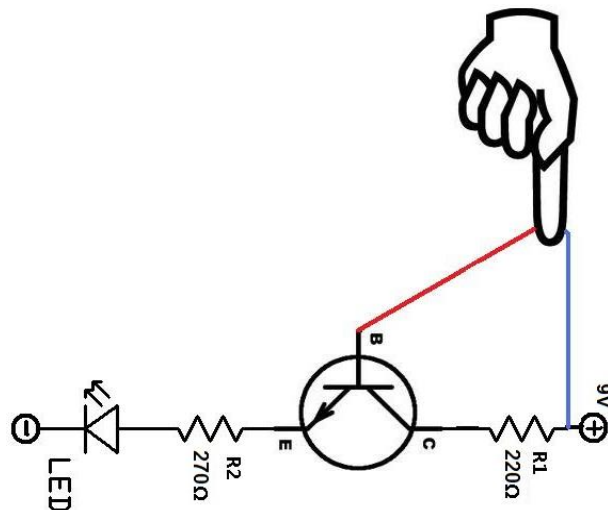
We can see that there is direct linkage between  $\frac{L_1}{L_1+L_2}$  and  $\frac{U_1}{U}$ , hence we can say  $\frac{L_1}{L_1+L_2} = \frac{U_1}{U}$

Hence, turns out the rheostat divides the voltage proportional to the length.

## Part 3

### Introduction

In this experiment a touch strip is included. The strip has only length, so the sensor is one directional. The goal of the experiment is to find the location of a touch on the sensor.



## Instruments

| Name             | Quantity | Properties                                  |
|------------------|----------|---------------------------------------------|
| Alligator Clips  | ~        | Type of wire with alligator interface       |
| Multi Meter      | 1        | Resolution of 0.001 V                       |
| Power Supply 15V | 1        | Generate AC/DC current through the circuit. |
| Conducting Tape  | 1        | Tape that has a layer of graphite           |

## Variables

| Name      | Type      | Resolution     |
|-----------|-----------|----------------|
| $L_1$     | Control   | 0.001 m        |
| $L_2$     | Variable  | 0.001 m        |
| $V_1$     | Measure   | 0.001 V, 0.01V |
| $V_{all}$ | Parameter | 0.01 V         |

## Errors

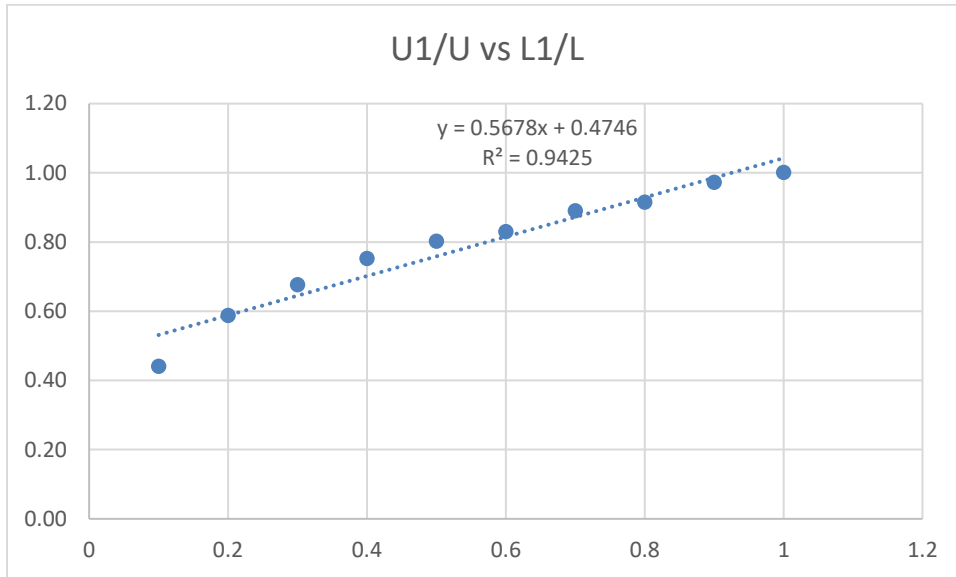
| Name                                     | Type       | Solution                                                                       |
|------------------------------------------|------------|--------------------------------------------------------------------------------|
| Resistance within wires                  | Systematic | Use wires with material that has low resistance                                |
| Voltmeter dash screen value oscillations | Random     | Try to hold still the terminals or just change the multi meter to a better one |
| Poor connections                         | Random     | Make connections more stable                                                   |
| Poor sensor coating                      | Random     | Coat the sensor more uniformly                                                 |

## Data

| L1   | L   | V1   | V | L1/L | U1/U |
|------|-----|------|---|------|------|
| 0.03 | 0.3 | 3.53 | 8 | 0.1  | 0.44 |
| 0.06 | 0.3 | 4.7  | 8 | 0.2  | 0.59 |
| 0.09 | 0.3 | 5.41 | 8 | 0.3  | 0.68 |
| 0.12 | 0.3 | 6.02 | 8 | 0.4  | 0.75 |
| 0.15 | 0.3 | 6.42 | 8 | 0.5  | 0.80 |

|      |     |      |   |     |      |
|------|-----|------|---|-----|------|
| 0.18 | 0.3 | 6.64 | 8 | 0.6 | 0.83 |
| 0.21 | 0.3 | 7.12 | 8 | 0.7 | 0.89 |
| 0.24 | 0.3 | 7.32 | 8 | 0.8 | 0.92 |
| 0.27 | 0.3 | 7.78 | 8 | 0.9 | 0.97 |
| 0.3  | 0.3 | 8.01 | 8 | 1   | 1.00 |

Plot



We have previously found an expression of  $\frac{L_1}{L} = \frac{U_1}{U}$

It is visible that the regression line has a slope of almost  $R^2 = 0.943$ , which is a strong value, but it could have been stronger, this implies that the formula works, however there might be some small errors or poor sensor coating quality.

The position of touch can be found by analytical expression of:

$$L(\text{position}) = U_1(\text{Voltmeter reading}) \times \frac{L(\text{Length of the sensor})}{U(\text{Total voltage in the sensor})}$$